

0.1 Experimental Evidence Overview

This section describes the experimental evidence that supports the analyses presented in this chapter. The evidence is organised into three categories: classification experiments, segmentation experiments, and mobile-deployment benchmarks. Each category is further subdivided into primary retained results and supplementary supporting evidence.

0.1.1 Classification Evidence

The primary classification evidence consists of machine-readable result files, configuration logs, and trained checkpoints produced under a fixed seed 42. The SDI-4 dataset, comprising 1,149 images across four classes (Healthy, BG, WSSV, WSSV_BG), was partitioned into training, validation, and test sets at an approximate 70:15:15 ratio (804/172/173). The final proposed classifier—YOLO26m-cls augmented with ASL-LDAM and SimAM-DCFR—achieved a Macro-F1 of 91.01%, accuracy of 91.33%, and Cohen’s Kappa of 0.8816 on the SDI-4 test partition. The cross-entropy (CE) baseline, which shares the same YOLO26m-cls backbone and training protocol without the custom attention and loss modifications, attained 89.02% Macro-F1 and 89.02% accuracy.

Supplementary classification evidence includes:

- **YOLO baseline screening:** A controlled comparison of 15 YOLO-cls variants (YOLOv8, YOLO11, YOLO26) at the nano, small, medium, large, and extra-large scales under the CE baseline protocol. The retained machine-readable CSV (yolo_baseline_comparison_seed42.csv) reports Macro-F1, accuracy, Cohen’s Kappa, parameter count, and model size for each variant.
- **TIMM baseline screening:** A separate benchmark of 17 classification architectures from the TIMM library (including ConvNeXt, MobileNet, EfficientNet, and lightweight variants) under the same SDI-4 partition and CE protocol. Results are recorded in timm_baseline_comparison_seed42.csv.
- **Component analysis:** Attention-benchmark logs from an RTX 4090 comparing the ASL-LDAM, SimAM-DCFR, and combined configurations against the CE baseline. These are available under the archived evidence: CVio_SDI4_ASL_LDAM_Attention_Benchmark_RTX4090.zip and the CE counterpart.
- **Additional-source evaluation:** The EXT-3-Original dataset (315 images, three classes) and the integrated SDI-4 + EXT-3-Original experiment, with partition-wise merged training, validation, and test splits. Retained results include class-level confusion matrices and source-stratified accuracy.
- **Kaggle T4 inference:** GPU-accelerated inference benchmarks recorded in CVio_Final_Proposed_T4_Seed42_FINAL.zip and CVio_Classification_Kaggle_T4x2_Evidence_2026-08-11.zip.

0.1.2 Segmentation Evidence

Segmentation experiments were conducted by team member Nguyen Van Phong. The primary retained evidence for the segmentation analysis includes:

- Baseline YOLO11n-seg results on the shrimp disease segmentation dataset, with masks for BG and WSSV classes (Healthy images are included as negative samples with no positive mask polygons).
- Attention-based segmentation variants and Fourier-domain segmentation experiments, evaluated under the same held-out test partition.
- Supplementary robustness and failure-mode analyses documenting known failure cases.

Detailed segmentation results are presented in Sections ?? through ?. The classification agent does not modify segmentation claims; it reproduces them from the accepted baseline evidence.

0.1.3 Mobile-Deployment Evidence

Mobile deployment evidence is preliminary and comprises TFLite conversion artifacts (FP32 and FP16 variants) and latency measurements obtained under controlled conditions. The retained artifacts include:

- CVio_SD14_Mobile20_seed42_2026-08-10.zip: TFLite delegates and inference logs.
- yolo26m_asl_ldam_simam_dcfr_fp32.tflite: FP32 TFLite model (approximately 39.52 MiB).
- yolo26m_asl_ldam_simam_dcfr_fp16.tflite: FP16 TFLite model (approximately 19.81 MiB).

Mobile evidence is discussed in Section ?. The TFLite results are treated as prototype evidence; they do not constitute clinical or real-time deployment guarantees.

0.1.4 Evidence Provenance and Limitations

All primary numerical claims in this chapter are traceable to machine-readable CSV files, configuration logs, or trained checkpoint hashes. The following hash values are recorded for the core SDI-4 checkpoints:

- Proposed SDI-4 checkpoint SHA-256: bf4d969465025e870a61be5e218dfc55b9b2e962dec416f220b1f130703fb1c7
- CE baseline SDI-4 checkpoint SHA-256: 4305e49158129c4a6acaa8fdaf7b5982a7f4d93cc233f11d17479e04b0e438c1

Where evidence is incomplete or where a numerical claim must be inferred from a visual source rather than a machine-readable record, the limitation is stated explicitly. No numerical claim in this chapter is fabricated or inferred from a plot when a machine-readable counterpart should exist.

0.2 Dataset Composition and Evaluation Partitions

This section describes the composition of the datasets used in the classification and segmentation experiments and the corresponding evaluation partitions.

0.2.1 SDI-4 Classification Dataset

The SDI-4 dataset is the primary evaluation benchmark for the classification task. It contains 1,149 RGB images of shrimp, sourced from the ShrimpDiseaseImageBD repository and augmented by the project team. The dataset is organised into four classes:

- **Healthy** (403 images): Shrimp specimens with no visible signs of disease.
- **BG** (198 images): Shrimp specimens exhibiting black gill disease.
- **WSSV** (328 images): Shrimp specimens infected with White Spot Syndrome Virus.
- **WSSV_BG** (220 images): Shrimp specimens exhibiting both WSSV and BG co-infection.

All images were resized to a uniform resolution of 224×224 pixels, consistent with the input requirements of the YOLO-cls architecture.

Partitioning Strategy

The dataset was partitioned using a fixed random seed of 42 to ensure reproducibility. The split was conducted at the image level using a stratified strategy that preserves the class distribution across partitions. The resulting partition sizes are:

$$\text{Train} = 804 \quad (70\%), \quad \text{Validation} = 172 \quad (15\%), \quad \text{Test} = 173 \quad (15\%).$$

The class-level distribution within each partition is reported in Table 1.

0.2.2 EXT-3-Original Dataset

The EXT-3-Original dataset consists of 315 RGB images sourced from an external collection, covering three classes: Healthy, BG, and WSSV. This dataset does not include the WSSV_BG co-infection class. The images were captured under different environmental conditions and resolutions from the SDI-4 dataset, providing a domain-shift evaluation scenario.

Table 1: Class distribution across SDI-4 partitions (seed 42).

Class	Train	Validation	Test	Total
Healthy	282	60	61	403
BG	139	30	29	198
WSSV	229	49	50	328
WSSV_BG	154	33	33	220
Total	804	172	173	1,149

The partition was fixed at:

$$\text{Train} = 221, \quad \text{Validation} = 47, \quad \text{Test} = 47.$$

0.2.3 Integrated SDI-4 + EXT-3-Original Dataset

To evaluate cross-source generalisation, the SDI-4 and EXT-3-Original datasets were merged at the partition level: each dataset was split independently using the same seed 42, and the corresponding partitions (train, validation, test) were concatenated. The resulting integrated dataset has:

$$\text{Train} = 1,025, \quad \text{Validation} = 219, \quad \text{Test} = 220.$$

The integrated test partition comprises images from both sources, with the WSSV_BG class appearing only in the SDI-4 subset. The test composition is reported in Table 2.

Table 2: Integrated test partition composition (SDI-4 + EXT-3-Original).

Class	SDI-4 subset	EXT-3 subset
Healthy	61	11
BG	29	16
WSSV	50	20
WSSV_BG	33	—
Total	173	47

0.2.4 Segmentation Dataset

The segmentation dataset is derived from the same ShrimpDiseaseImageBD source. It contains images with pixel-level polygon annotations for two positive classes: BG and WSSV. Healthy images from the original source are retained in the segmentation training set as negative samples (i.e., images with no positive mask polygons). These images participate in the optimisation process by contributing zero-valued loss for the mask head, but they do not introduce positive mask supervision for the Healthy class.

The segmentation partition sizes and class distribution are documented in the segmentation experimental records (Section ??).